

Introduction

- Indoor environments pose significant challenges for autonomous navigation
 - GPS is unavailable indoors, and wireless signals such as Wi-Fi are often unreliable due to interference and multipath effects
 - This makes accurate indoor localization difficult using conventional methods
- Limitations and opportunities in indoor navigation
 - LiDAR-based systems offer high accuracy but are costly and increase system complexity
 - Indoor lighting conditions are relatively controlled, making vision-based approaches more practical and cost-effective

Introduction

- The project focuses on a hotel-like indoor corridor environment. Such environments contain repetitive and visually similar structures, which can cause ambiguity in visual localization.
- Use of navigation beacons
 - ArUco markers are used as navigation beacons to uniquely identify locations and resolve visual ambiguity
 - These markers are low-cost, easy to detect, and suitable for real-time indoor navigation.

Problem Definition

- Optimize the placement and number of ArUco tags to ensure reliable indoor robot navigation with minimal infrastructure.

Objectives

- To calibrate the camera for visual marker detection and robot navigation
- To generate a placement strategy and draw inferences for optimizing the placement of the Aruco markers
- To develop a multi-stage system using feasibility analysis, visibility matrix, deterministic and greedy algorithm-based placement
- To validate the proposed algorithm in a structured corridor in simulation and in real-time

Relation with SDG

SDG 9: Industry, Innovation, & Infrastructure

Affordable Automation: Creates a low-cost, GPS-independent navigation system for indoor industrial environments.

SDG 11: Sustainable Cities & Communities

Smart Infrastructure: Powers autonomous service robots to improve efficiency in public spaces like hospitals and airports.

SDG 12: Responsible Consumption & Production

Resource Efficiency: Minimizes energy waste and hardware usage through optimized pathfinding and tag placement.



Response to the Reviewer Comments -Review 2

Provide the detailed response to the reviewer comments (Review 2)

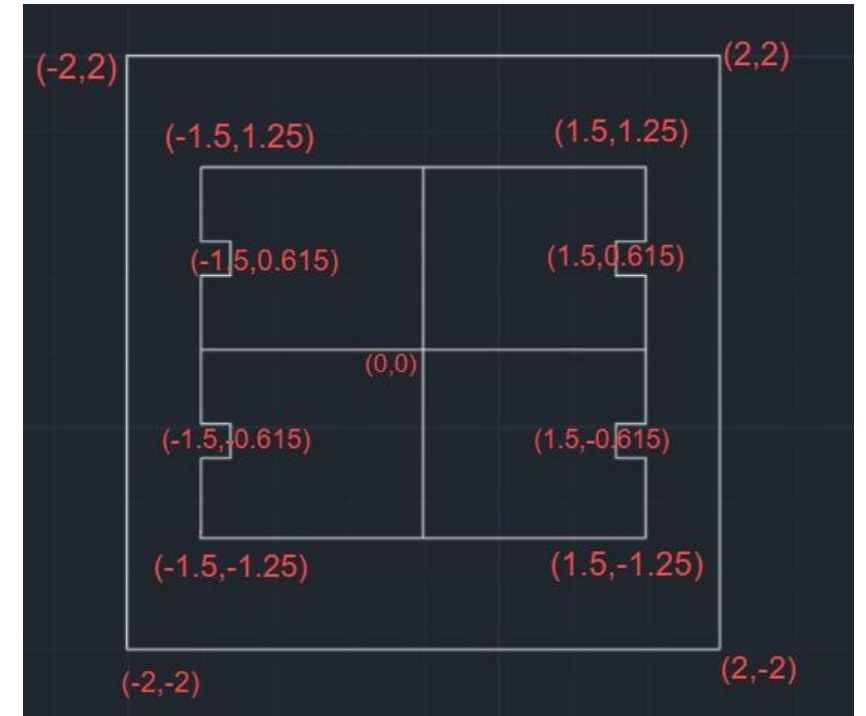
- Comment #1: Results need to be validated
- Response #1: Hardware demonstration helps validate the simulated results
- Comment #2: Clarity in terminology and structured presentation required
- Response #2: The presentation has been revised to improve clarity and structure. Parameters are defined before use.

Methodology – Environmental Setup

The environment includes functional areas such as doors and junctions, which are prone to localization errors. Corridor width is measured to define lateral visibility and the required bearing angles for detecting junctions before and after turns.

The beacons are categorized into three types based on placement locations as follows:

- Functional Tags(F) – placed at doors or task areas
- Junction Tags(J) – placed at intersections
- Corridor Tags(C) – for straight-line navigation.



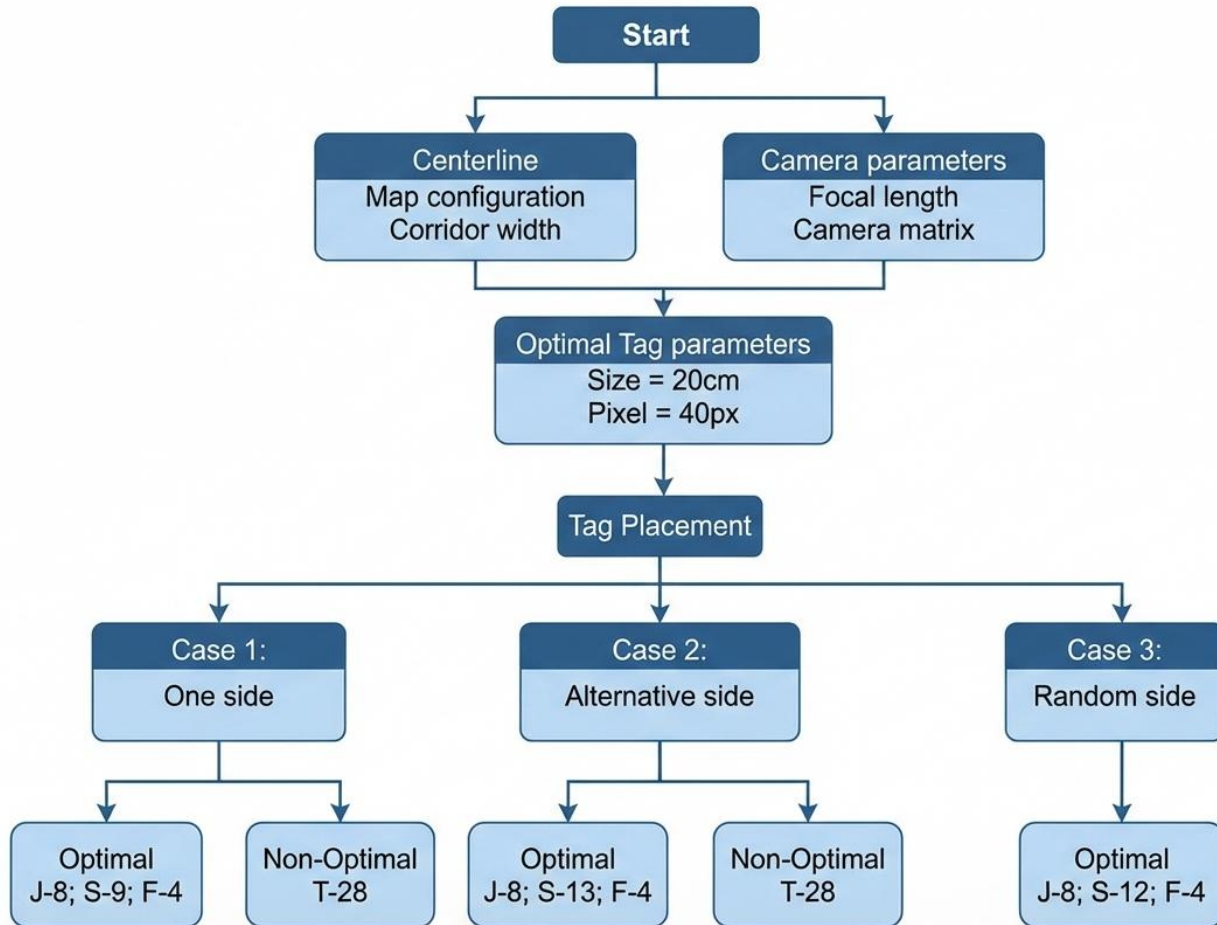
AutoCad drawing of the environment marked with coordinates

Methodology – Placement strategy

Visibility constraints are derived using the Environment and Camera parameters and are used as baseline constraints for manual placement technique. The cases are analyzed under Optimal and Non-Optimal configurations.

- Non-optimal Cases – Tags are placed uniformly with an interval of 25cm on single-sided walls and alternating sides of the corridor walls.
- Optimal Cases – Deployment follows specific visibility constraints, testing configurations – one side, alternate side, and a randomised optimal placement;

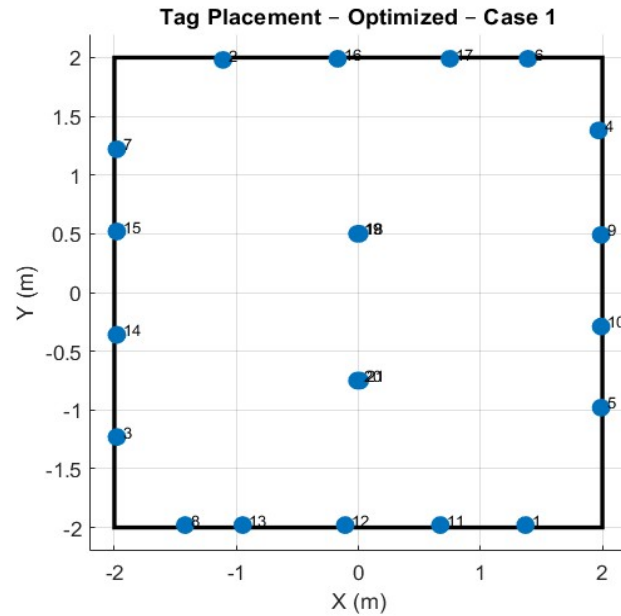
Methodology – Placement strategy



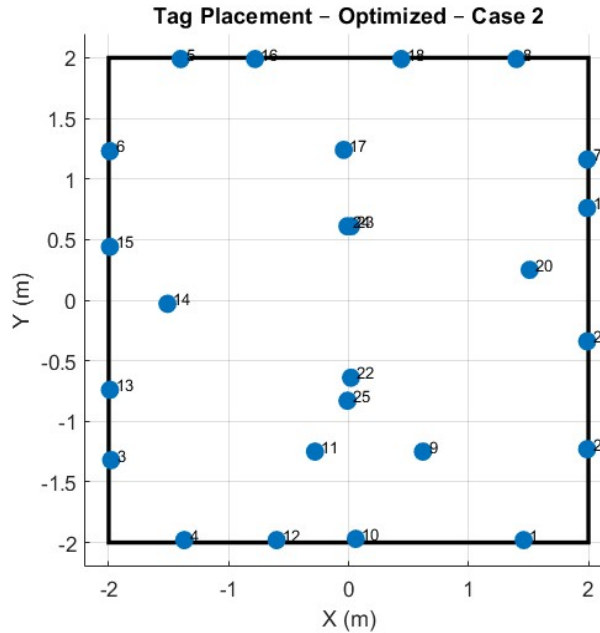
J- Junctional Tags
S- Straight Tags
F- Functional Tags
T- Total Tags

Methodology for optimizing tag placement in a robot's workspace to maximize localization availability across different corridor configurations.

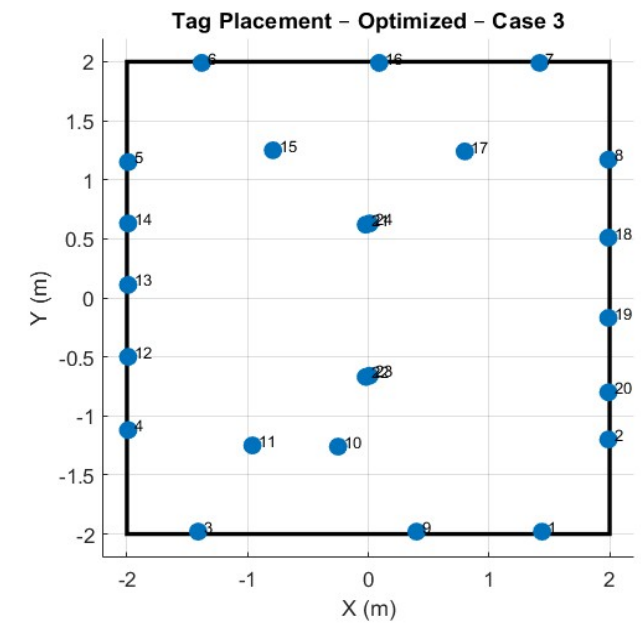
Methodology - Manual Tag Placement



Tags Placed - 21



Tags Placed - 25



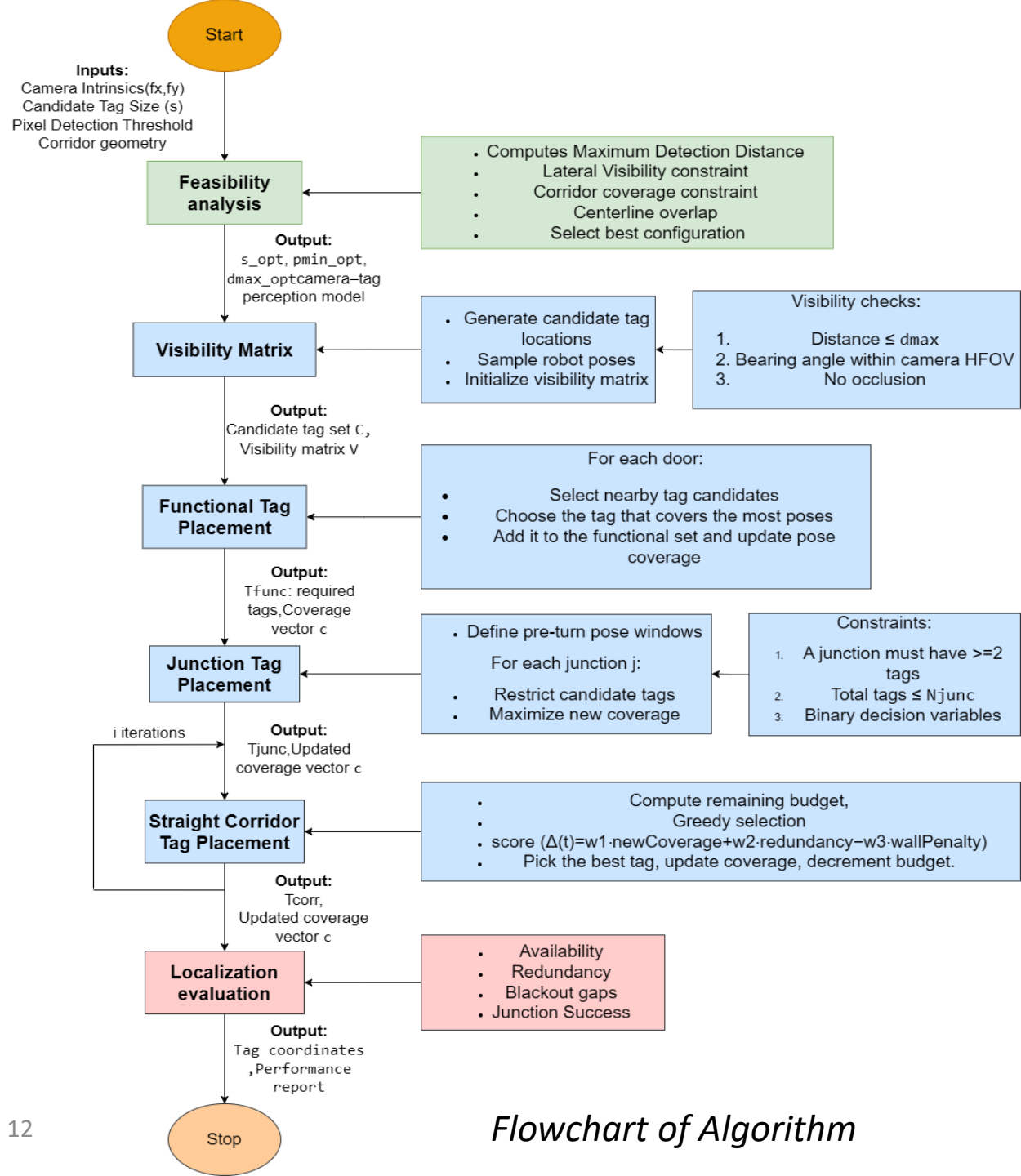
Tags Placed - 24

MATLAB-generated plot of three optimal cases of manual marker placement.

Methodology - Manual Tag Placement

Inferences:

- Tag size must be optimized: Small tags reduce localization accuracy, while large tags overcrowd the camera FOV.
- Uniform placement provides high redundancy ($>40\%$), improving robustness against occlusion, but increases infrastructure cost and deployment complexity.
- Junction and functional tags are essential for reliable navigation.
- Pre- and post-junction visibility improves re-localization and reduces failures.
- Alternating placement suits wide corridors; single-side placement is adequate for narrow corridors.

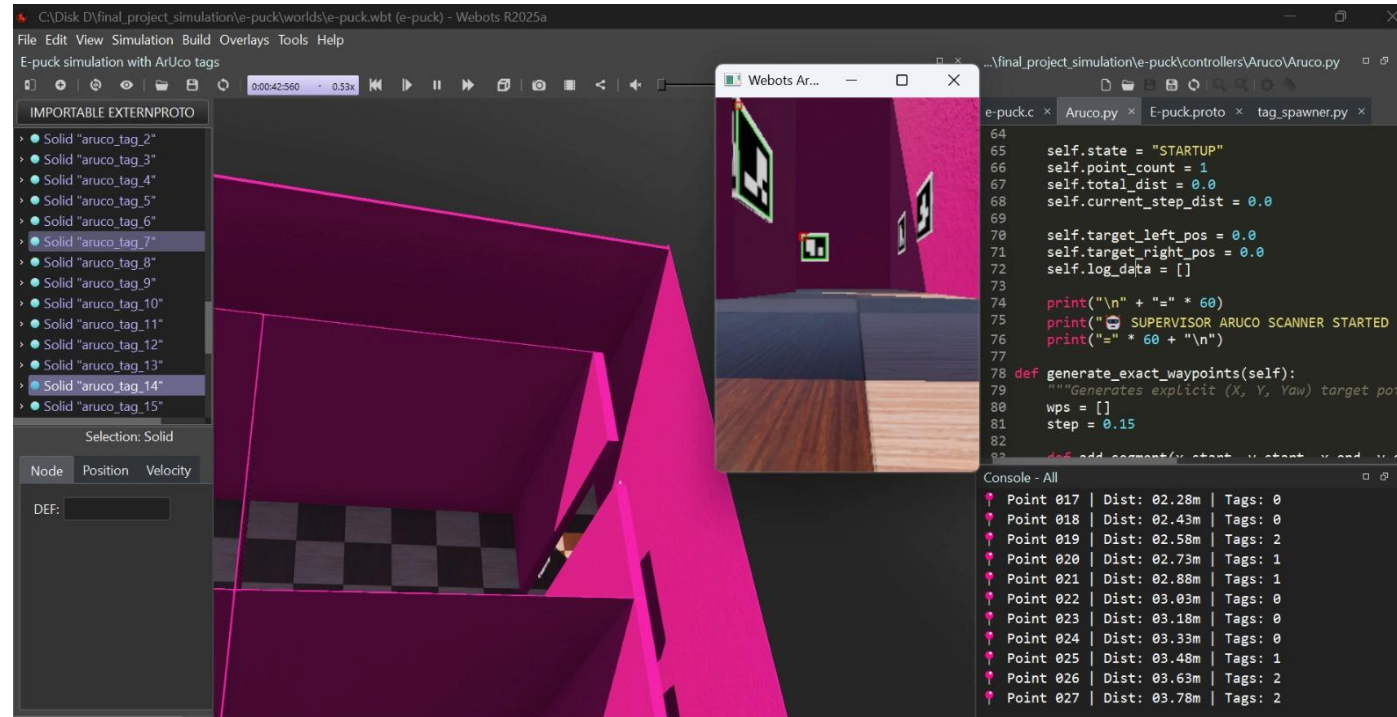


Methodology - Algorithm

A multi-stage optimization algorithm is designed to optimise the placement and number of ArUco tags to ensure reliable indoor robot navigation with minimal infrastructure.

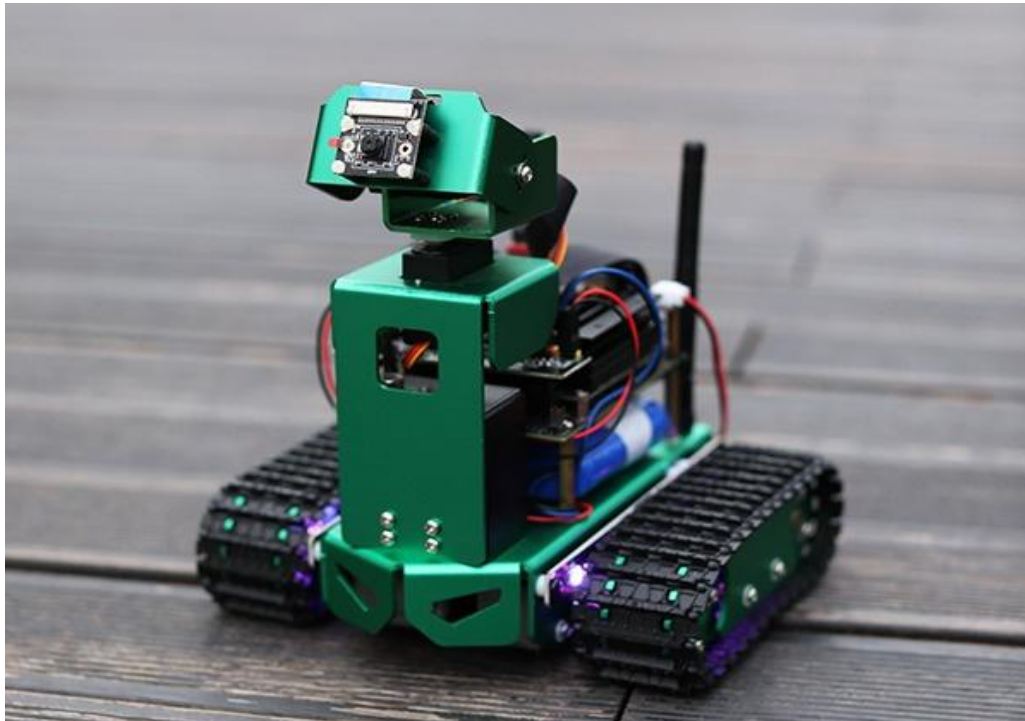
Minimal infrastructure refers to the trade-off between reduction of Aruco tags while increasing the localization availability

Methodology – Simulation environment



Simulation environment in webots

Hardware Implementation:



JETBOT

Key Components:

- NVIDIA Jetson Nano
- OLED Display
- Two-wheel differential drive motor system
- Dual-Band Wifi
- Yahboom IMX219HD Camera

Parameter	Specification
Sensor	8MP IMX219 CMOS (1/4 inch)
Max Resolution	3280 × 2464
Field of View (FOV)	160° (Wide-Angle)
Focal Length	3.15 mm
Aperture	2.35
Distortion	< 14.3%

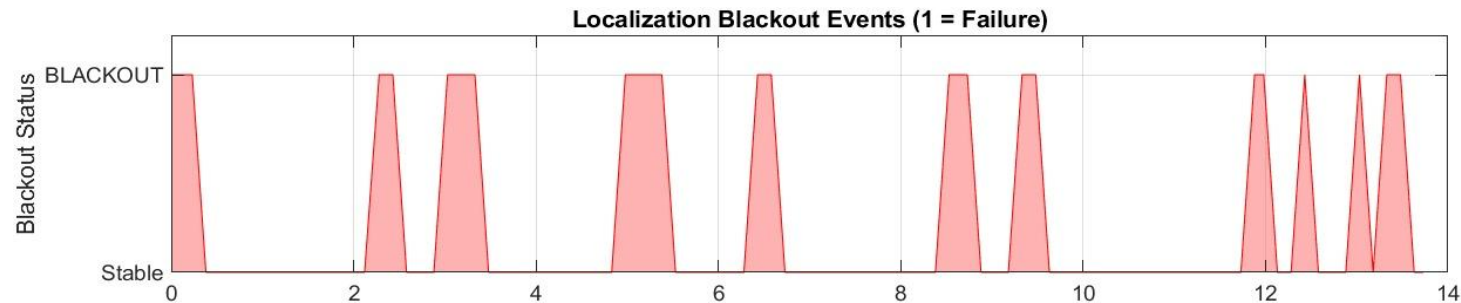
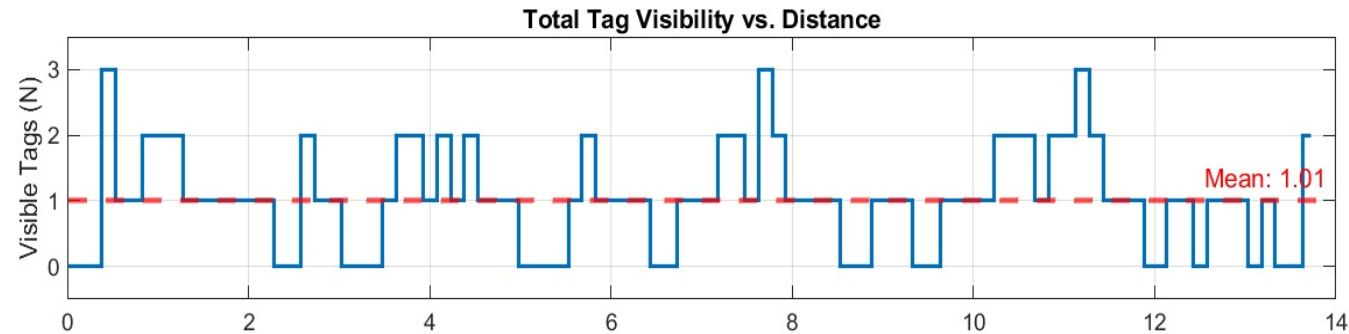
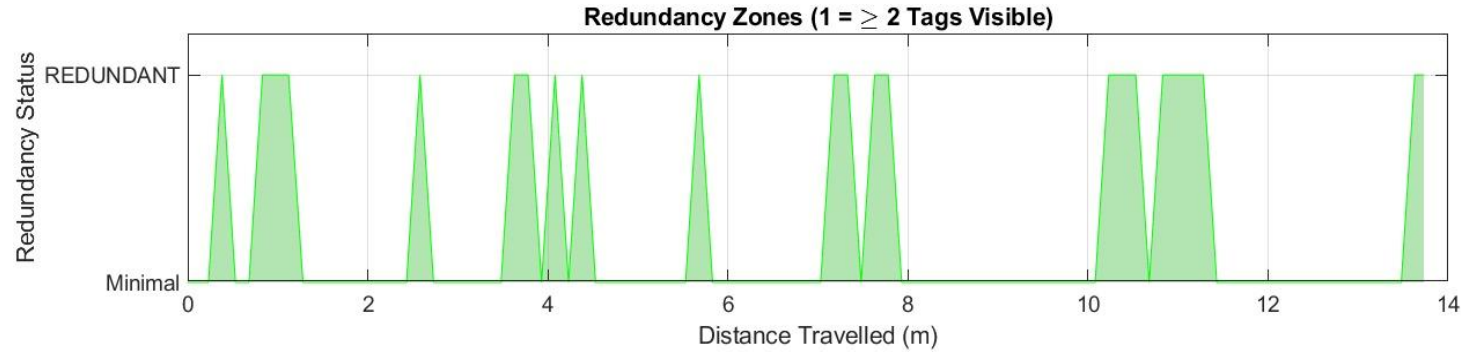
Implementation:



Environment setup:

The environment consists of a 4x4 outer boundary and a 4x4 inner boundary to make up a corridor like setup as showed in the simulated model

Key Results – Simulation result for algorithm-based Placement



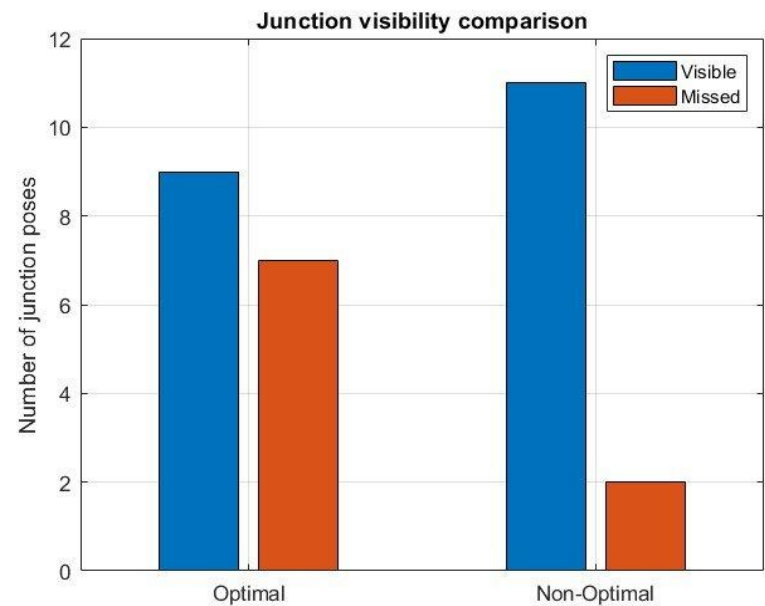
Analyses an indoor robot's ArUco tag localization performance by correlating total tag visibility with specific intervals of navigation blackout and redundant sensor coverage across a robot path.

Key Results – Simulation result for algorithm-based Placement

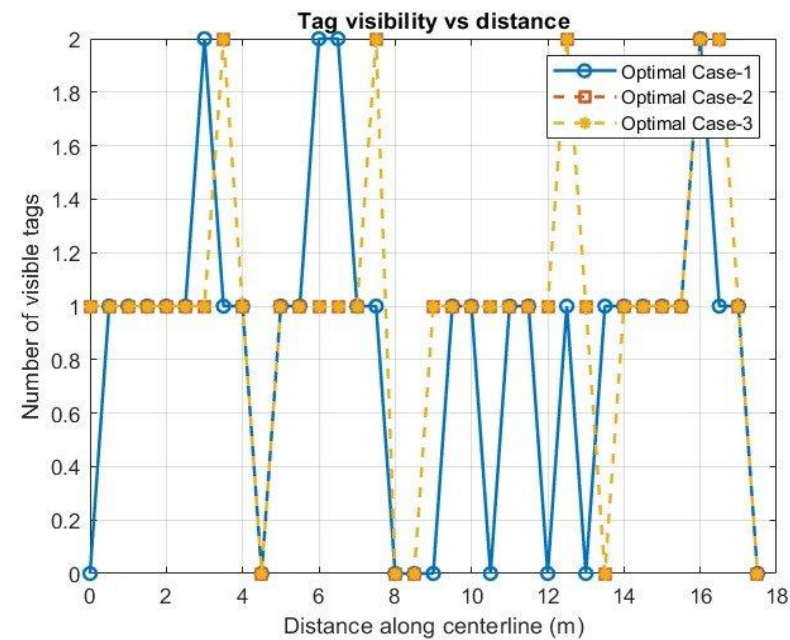
Table 5 - Performance Summary

Metric	Value
Total Distance Travelled	13.73 m
Mean Tags Visible	1.011
Max Tags Observed	3
Localization Reliability (Path with ≥ 1 Tag)	84.58%
Blackout Rate (Failure) (Path with 0 Tags)	16.32%
Redundancy Rate (Optimal) (Path with ≥ 2 Tags)	24.21%

Key Results – Simulation Result Manual Tag Placement

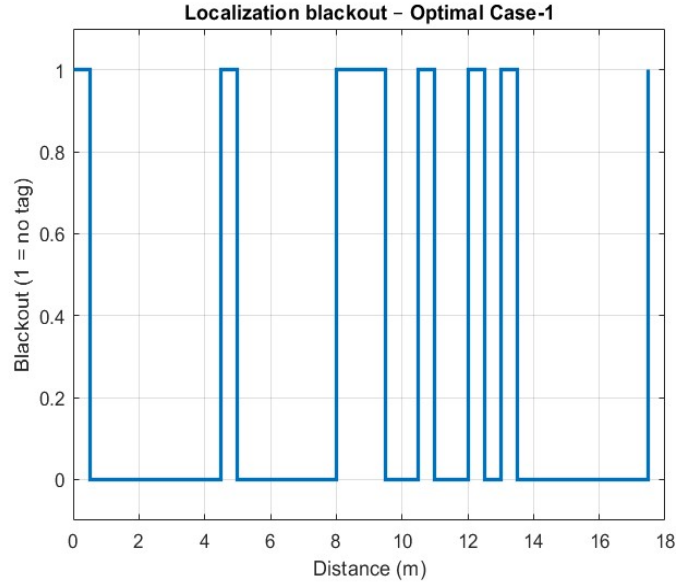


A grouped bar chart comparing the number of visible versus missed junction poses for optimal and non-optimal tag placement configurations.

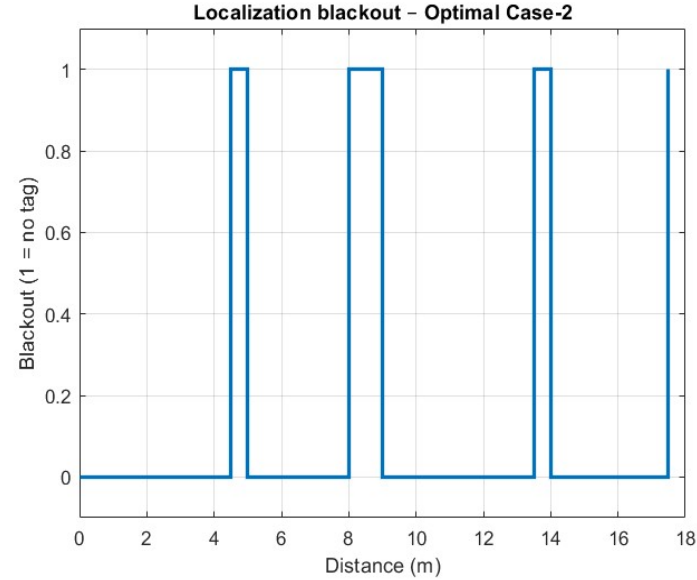


A collection of MATLAB-generated charts, including a grouped bar graph comparing visibility and a line plot tracking the number of visible tags over distance for three optimal placement cases.

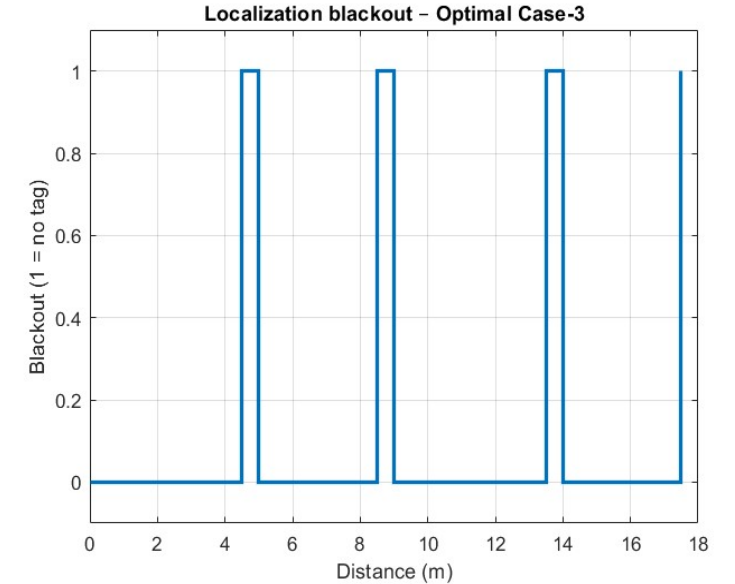
Key Results – Simulation result Manual Tag Placement



Case 1 of Blackout



Case 2 of Blackout



Case 3 of Blackout

Blackout – Locations where tags are not visible

Key Results – Simulation result Manual Tag Placement

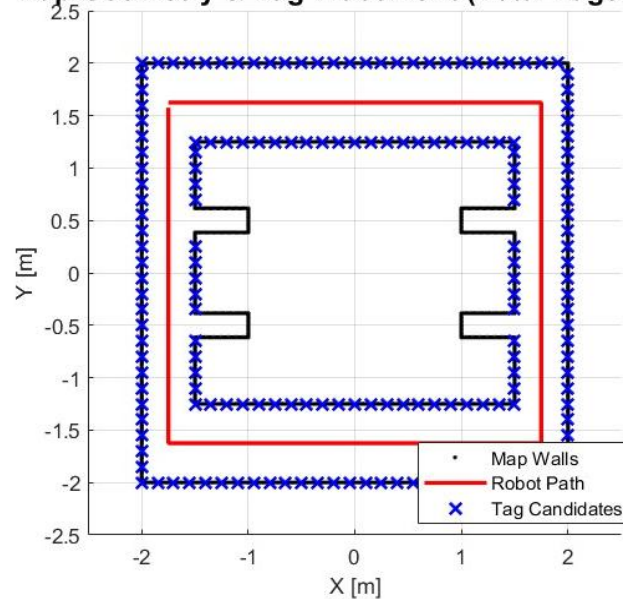
Table 1 – Results for Manual Tag Placement

Case	Total tag coverage (%)	J-tag coverage (%)	Redundancy (%)	Missed J-tags
Optimal C1	75	56.25	11.11	7
Optimal C2	86.11	68.75	13.88	5
Optimal C3	88.88	75	11.11	4
Non-Opt. C1	77.77	68.75	44.44	2
Non-Opt. C2	77.77	68.75	30.55	2

- Optimal_Case3 achieves the highest Coverage (88.89%) and J-tag Coverage (75%), indicating consistently reliable tag availability.
- It also maintains low Redundancy (11.11%), the smallest MaxGap (0.5 m), and fewer Missed Junctions (4), providing the best balance between efficiency and coverage quality among all cases.

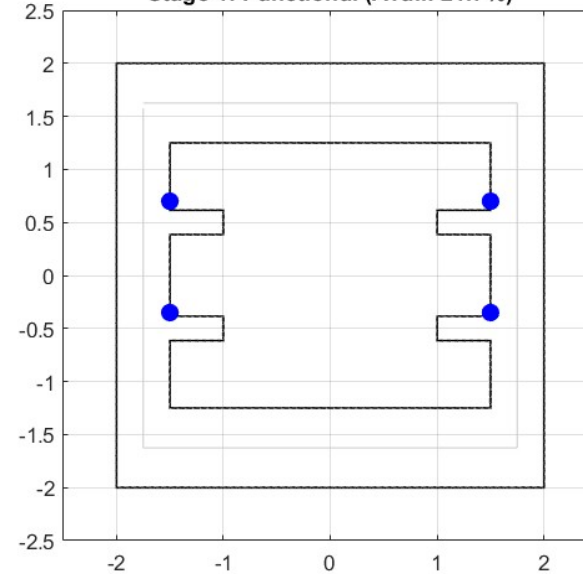
Key Results – MATLAB result for algorithm-based Placement

Map Geometry & Tag Placement (Total Tags: 178)



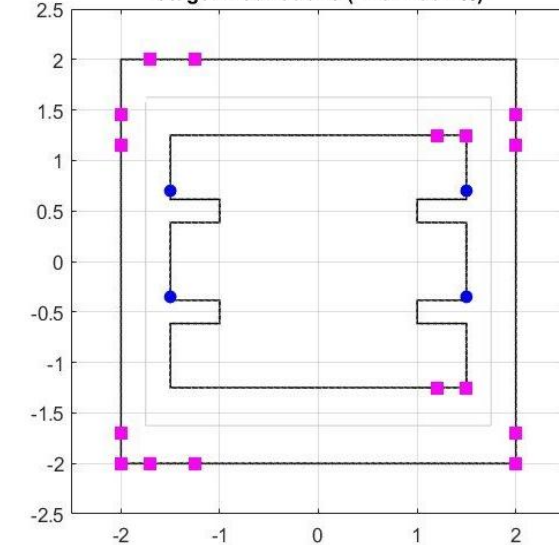
A map visualizing the indoor geometry with Map Walls - black, the Robot Path in red, and 178 Tag Candidates marked as blue 'x' symbols for your navigation strategy.

Stage 1: Functional (Aval: 21.7%)



A refined version of the map showing only 4 Functional tags that provide a 21.7% localization availability along the robot's intended path.

Stage 2: Junctions (Aval: 60.1%)



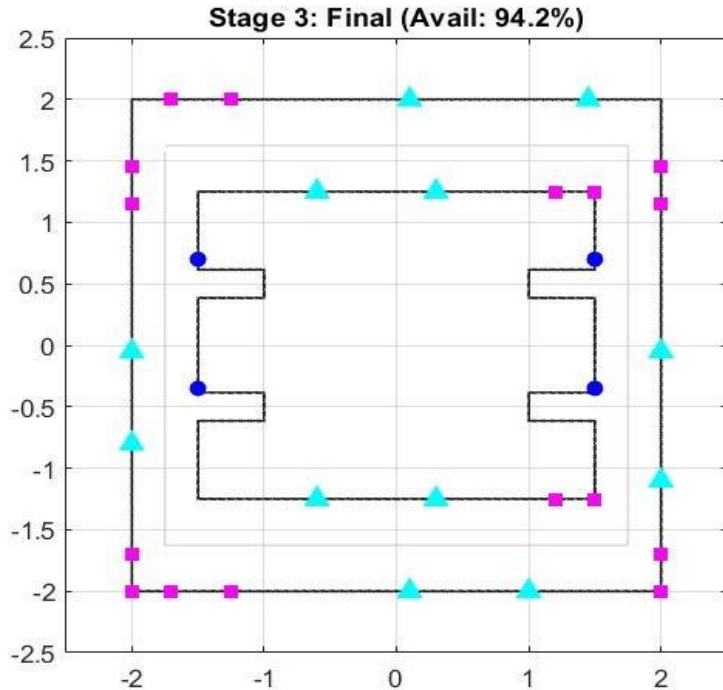
Adds tags at corners and intersections (magenta squares), increasing localization coverage to 60.1%.

Key Results – MatLab result for algorithm based Placement

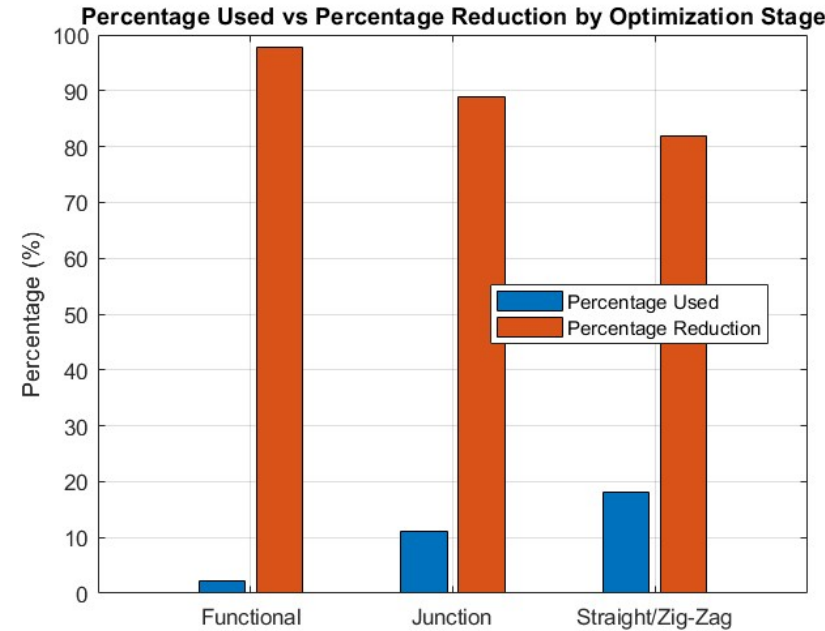
Table 2 – Coverage improvement across iterations

Iter	Coverage (%)	No tag covered frames	Candidate tags	Selected Tags
1	64.49	49	158	21
2	68.84	43	157	22
3	73.19	37	156	23
4	77.54	31	155	24
5	81.88	25	154	25
6	84.78	21	153	26
7	86.96	18	152	27
8	89.13	15	151	28
9	90.58	13	150	29
10	92.03	11	149	30
11	93.48	9	148	31
12	94.20	8	147	32

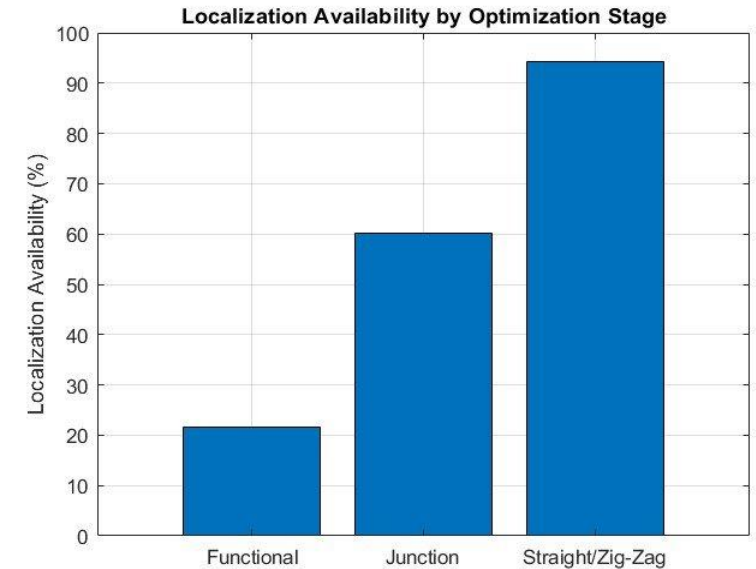
Key Results – MATLAB result for algorithm-based Placement



Incorporates additional markers at corridors (cyan triangles) to reach a near-total coverage of 94.2% for reliable navigation.

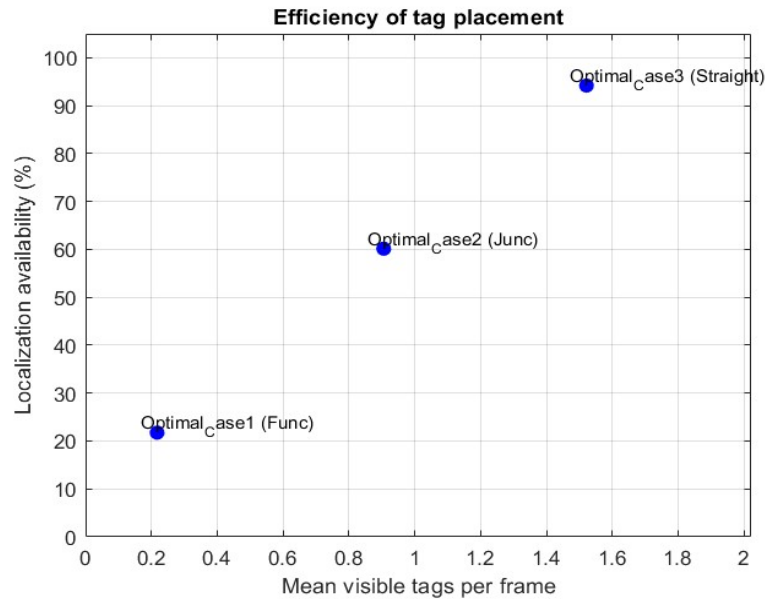


Comparing the Percentage Used and Percentage Reduction across three optimization stages: Functional, Junction, and Straight/Zig-Zag.

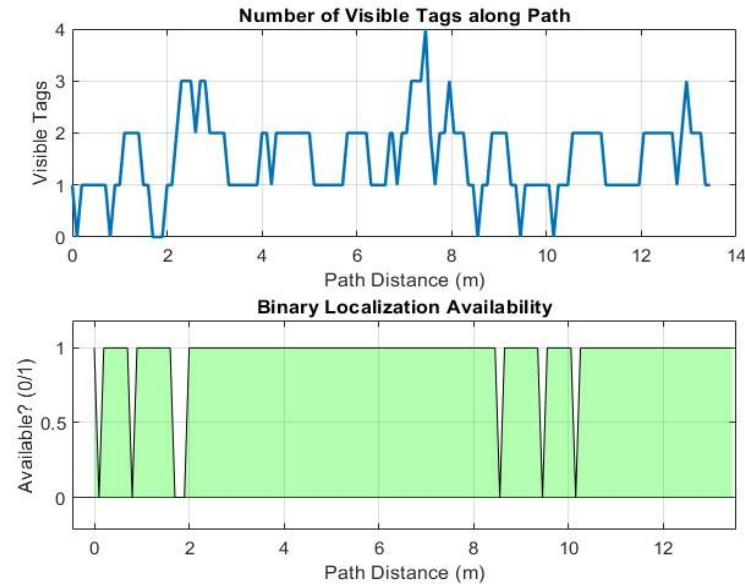


Showing the increase in Localization Availability (%), i.e. the percentage of frames with at-least one tag is covered, as the optimization progresses through Functional, Junction, and Straight stages.

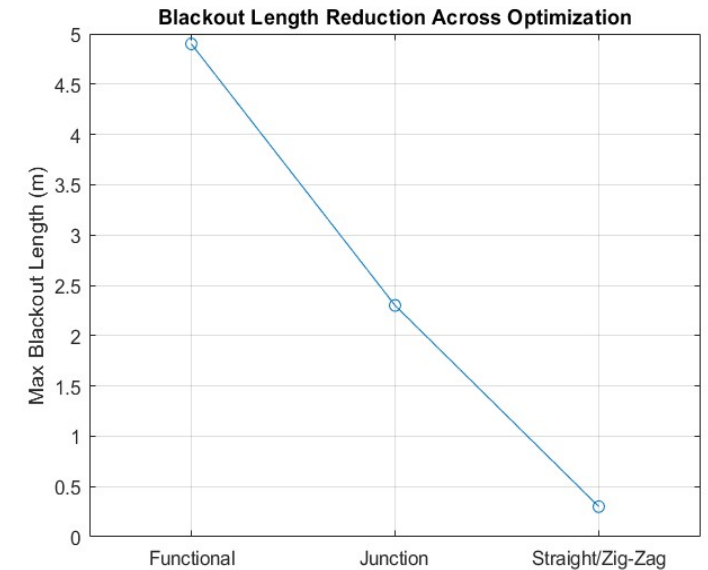
Key Results – MATLAB result for algorithm-based Placement



This illustrates the direct correlation between the number of visible ArUco tags and localization availability.



This dual-plot visualization tracks the number of visible ArUco tags and the resulting binary localization availability along a robot path.



Shows a significant decrease in maximum blackout length (m) across three optimization stages: Functional, Junction, and Straight/Zig-Zag.

Key Results – MatLab result for algorithm-based Placement

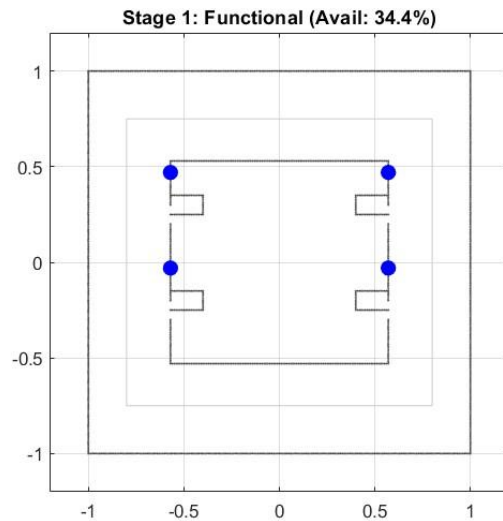
Table 3 – Summary Metrics

Stage	Tags Used(%)	No.Tags Used	Total Candidates	Reduction(%)	Localization Availability(%)	Total Blackout Length(m)
Functional Tags	2.2	4	178	97.8	21.7	4.9
Junction Tags	11.2	20	178	88.8	60.1	2.3
Straight	18	32	178	82	94.2	0.3

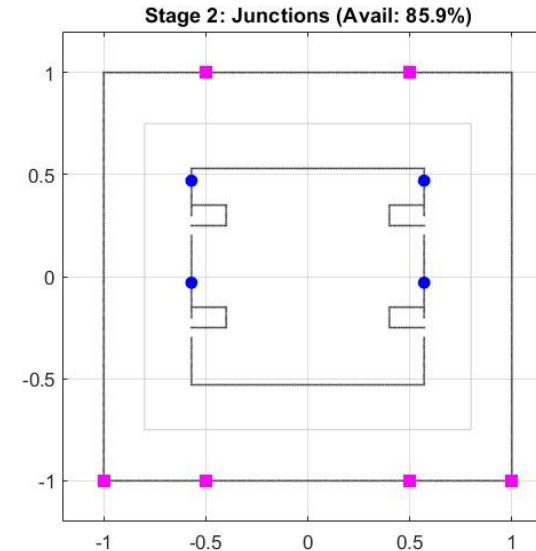
Table 4 - Final Summary Metrics

Tags Used(%)	No. Tags Used	Total Candidates	Reduction(%)	Localization Availability(%)	Blackout Length(m)
18	32	178	82	94.2	0.3

Hardware Results:

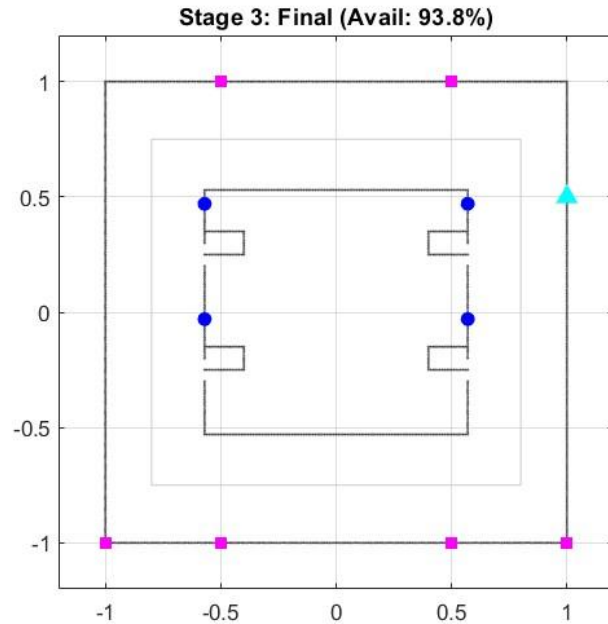


Metric	Value
Tags Used	4 of 118
Percentage Used	3.4%
Percentage Reduction	96.6%
Localization Availability	34.4%
Max Blackout Length	2.20 m

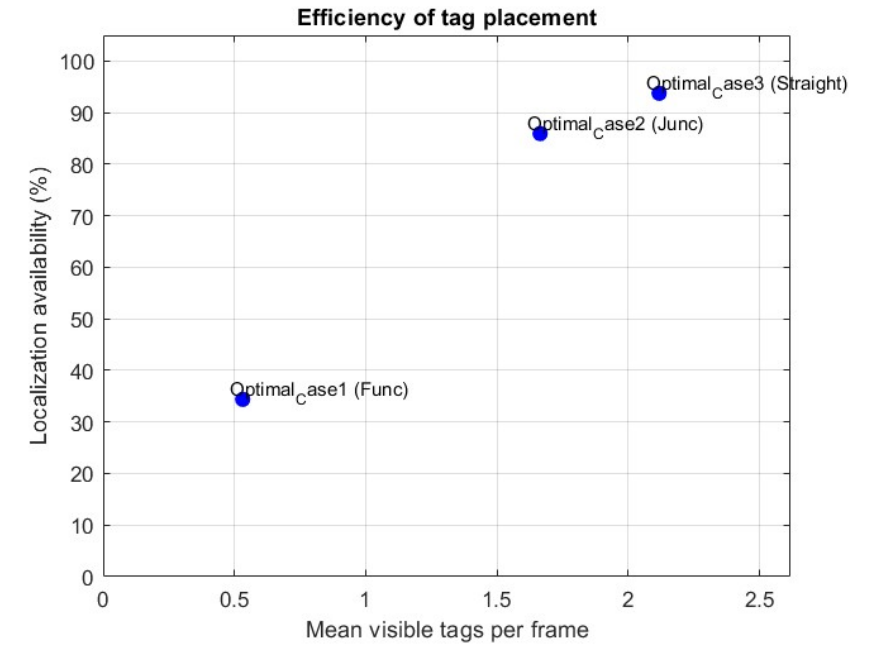


Metric	Value
Tags Used	10 of 118
Percentage Used	8.5%
Percentage Reduction	91.5%
Localization Availability	89.9%
Max Blackout Length	0.70 m

Results:

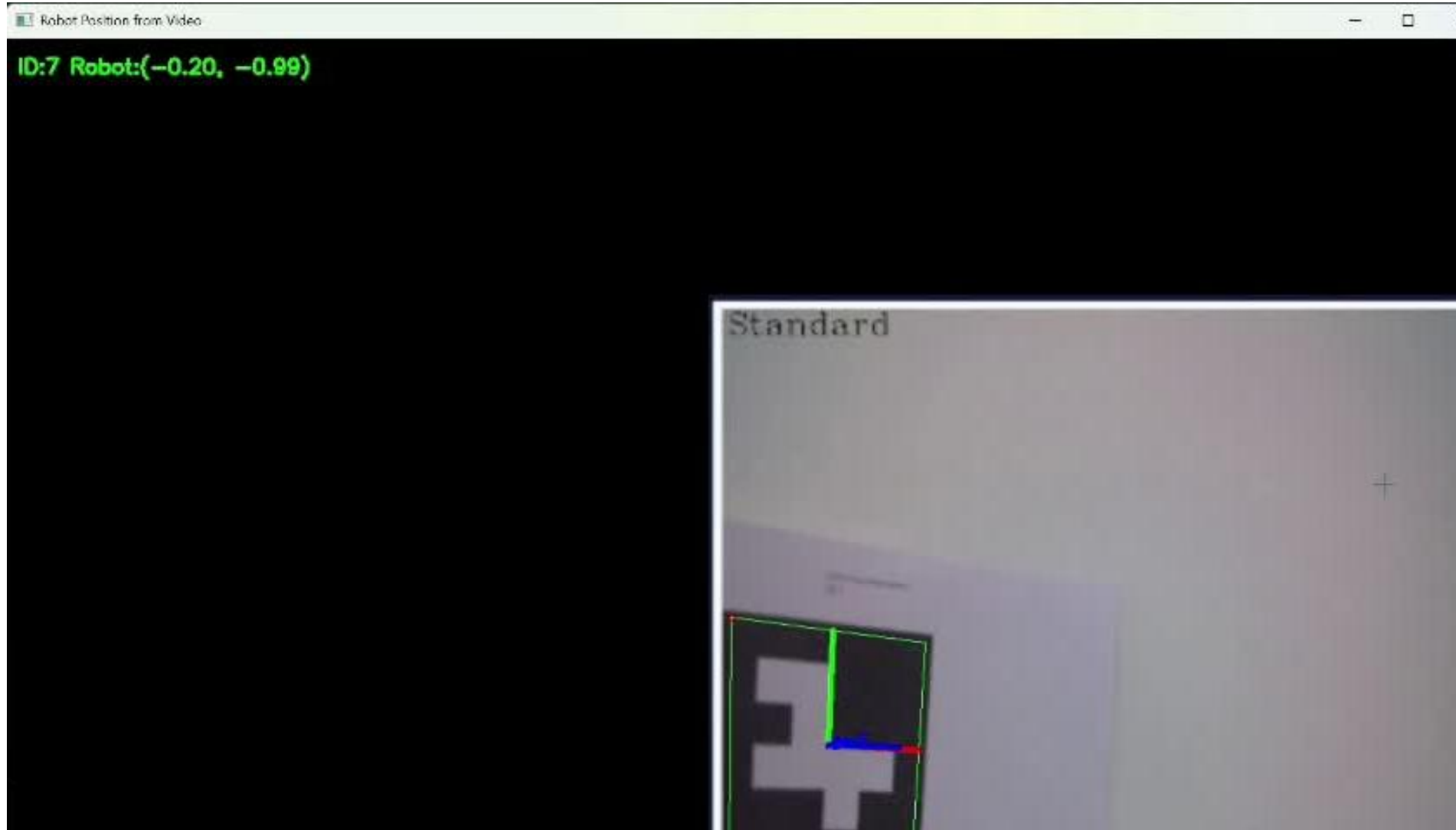


Iteration	Coverage (%)	Candidates Remaining	Total Tags
1	93.75	108	11

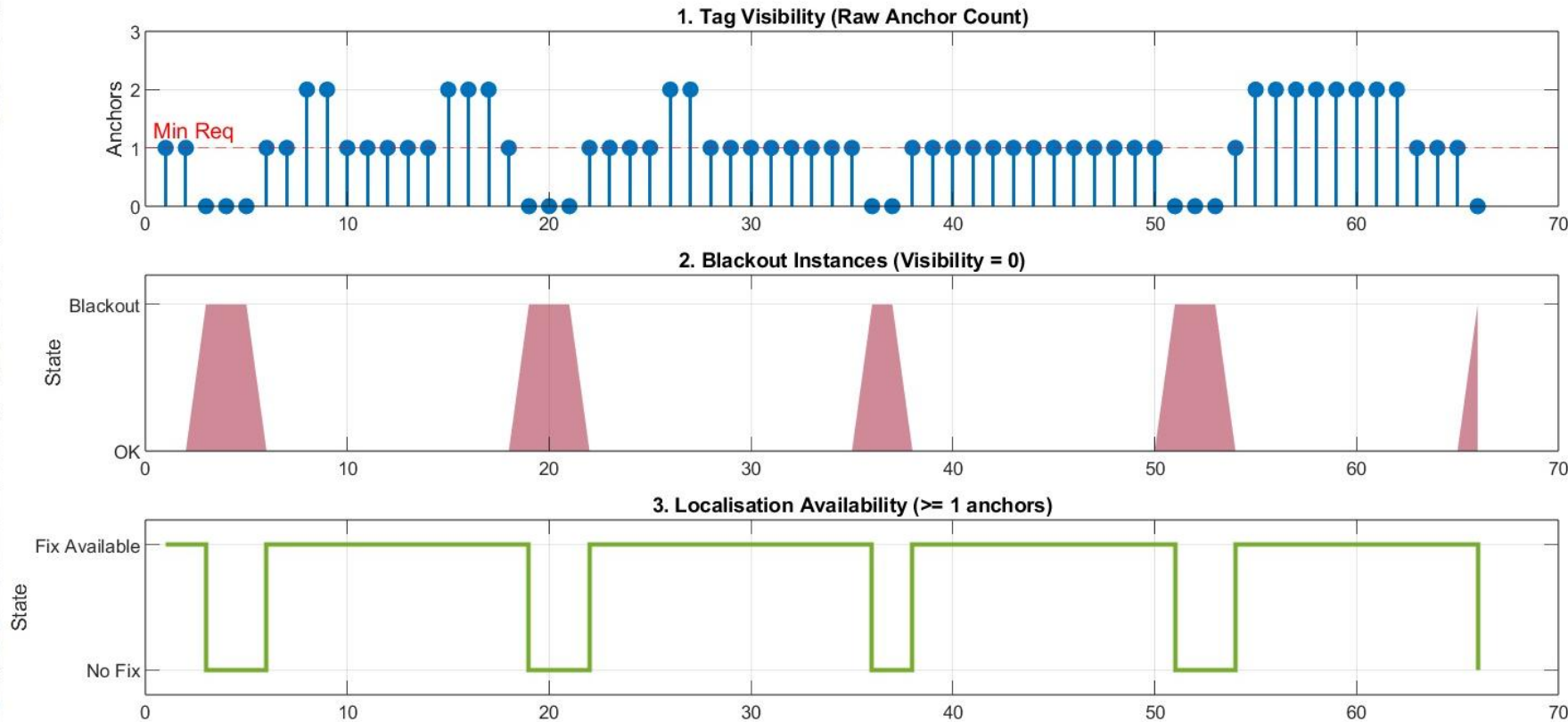


This illustrates the direct correlation between the number of visible ArUco tags and localization availability.

Key Results:



Key Results – Hardware output



Overall Statistics:

- Total Frames: 66
- Blackout Percentage: 18.18%
- Availability Percentage: 81.82%
- Average Efficiency: 81.79%

- 1.Tag visibility: No. of tags visible in the particular frame
- 2.Blackout Instances: Regions where no tags were visible
- 3.Localization Availability: Regions where at least one tag was visible

Conclusion

- The system achieved a 94.2% localization availability in both MATLAB and Webots simulations.
- Optimal Placement Strategy: Utilizing a visibility-aware greedy selection technique proved more effective than uniform marker distribution, which was found to be inefficient and redundant.
- Junction Safety: By mandating dual-tag placement at every intersection, the system achieved a 100% pre-turn success rate, resolving a common research gap where robots often lose localization during high-speed turns.
- Cost-Effectiveness: The use of passive ArUco markers provides a low-cost, computationally efficient alternative to expensive LiDAR-based systems, making it suitable for resource-constrained hardware like the NVIDIA Jetson Nano.
- Validation: The results were consistent across theoretical modeling and physical implementation on a Yahboom Jetbot.



References

1. Alghamdi, M., Al-Marakeby, A. and Abdel-Mageid, S. (2025) 'Mobile robot navigation based on artificial markers: A systematic mapping study', *Journal of Robotics and Mechatronics*
2. Mantha, B.R.K. and García de Soto, B. (2022) 'Investigating the fiducial marker network characteristics for autonomous mobile indoor robot navigation using ROS and Gazebo', *Journal of Construction Engineering and Management*
3. Beinhofer, M., Müller, J., Krause, A. and Burgard, W. (2013) 'Robust landmark selection for mobile robot navigation', in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

Thank You